

llmXive Automated Review of Beyond Scalar Rewards by Internalizing Reasoning into Score Distributions

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with cited sources.

Citation Accuracy and Overstatement: In Table 1 (lines 130-165 of `main-llmxive.tex`), the row for **HPSv3** marks “Score Distribution” as supported (`\tabyes`). The citation provided is `ma2025hpsv3widespectrumhumanpreference`. While HPSv3 is a strong scalar reward model, the claim that it explicitly models a *distribution* of scores (as opposed to a single scalar value) requires verification against the cited source. If HPSv3 outputs a single scalar, this table entry is factually incorrect and misrepresents the baseline. Similarly, the claim that **RewardDance** uses chains distilled from “Qwen-3.6-Max” (Section 4.1, line 635) is a very specific implementation detail. If the original `wu2025rewarddancerewardscalingvisual` paper does not specify this exact model version or the distillation source, the authors should generalize the claim to avoid attributing specific engineering choices to the cited work that may not be present.

Claim Strength vs. Evidence: The abstract and Section 4.1 claim the 9B student “closely matches” the 27B teacher (88.6% vs. 89.6% HPA). While the absolute difference is small (1.0%), in the context of reward modeling where baselines are often 70-80%, a 1% gap at the high end can be statistically significant. The text asserts this match without referencing statistical significance tests (e.g., p-values) or confidence intervals, which are typically required to support a claim of “closely matching” performance between models of vastly different sizes. The figures (e.g., `human_preference_accuracy_curve.pdf`) are not visible, but the text should qualify this claim (e.g., “statistically comparable” or “within X%”) if such analysis is not provided.

Methodological Attribution: The description of the **GRPO** baseline in Section 4.1 states it “computes rewards from the mean of the predicted score distribution.” The citation `deepseek-math` refers to the original GRPO paper, which typically uses scalar rewards derived from a reward model or a rule-based function. If the authors’ implementation of GRPO specifically uses a score-distribution mean (as opposed to a scalar token), this is a modification of the baseline. The text should clarify if this is a standard GRPO implementation or a specific adaptation by the authors to ensure the comparison is fair and the citation accurately reflects the baseline’s standard definition.

Overall, the paper makes strong claims about the capabilities of baselines and the performance of the proposed method. Minor revisions are needed to ensure that table entries for baselines strictly match the cited literature and that performance comparisons are qualified appropriately.

Action items.

- **[writing]** In Table 1 (`main-llmxive.tex`), the entry for ‘HPSv3’ claims it supports ‘Score Distribution’ (`\tabyes`). However, the cited work (`ma2025hpsv3widespectrumhumanpreference`) is primarily known for a scalar Human Preference Score. Verify if HPSv3 explicitly outputs a distribution or if this claim overstates the cited source’s capabilities.
- **[writing]** The abstract and Section 4.1 claim the 9B RISD student reaches 88.6% HPA, ‘closely matching’ the 27B teacher (89.6%). While the absolute difference is 1.0%, the relative gap in the 9B baseline (SFT 74.6%) is significant. Ensure the claim of ‘closely matching’ is supported by statistical significance testing or error bars in the figures, as a 1% gap in high-accuracy regimes can be non-trivial.

- **[writing]** Section 4.2 states that RewardDance ‘uses post-hoc pseudo reasoning chains distilled from Qwen-3.6-Max’. The citation (wu2025rewardddancerewardscalingvisual) should be checked to confirm if the reasoning chains are explicitly distilled from a specific ‘Qwen-3.6-Max’ model or if this is a specific implementation detail of the authors’ reproduction not present in the original paper.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure displays training curves clearly with a legend, but the caption is unpolished with raw citation keys and describes a ‘Right’ panel that is missing from the rendered image.

Figure 2

The figure displays clear learning curves with error bands, but the caption is too brief to explain the plot’s content, and the x-axis tick labels are illegible due to overlapping text.

Figure 3

The figure presents a learning curve with two overlapping lines but fails to identify what these lines represent due to a missing legend and an insufficient caption, rendering the data meaningless.

Figure 4

The figure fails to provide a valid qualitative comparison because it depicts two different models (subjects) rather than the same subject generated by different methods. Additionally, the caption contains a missing term (‘-guided optimization’) that obscures the method being evaluated.

Action items.

- **[writing]** Figure 1 caption: contains raw citation keys (‘wu2025rewardddancerewardscalingvisual’, ‘deepseek-math’) and mentions a ‘Right’ panel that is not present in the rendered image.
- **[science]** Figure 1: the legend lists ‘27B-RewardDance’ and ‘9B-SFT’ as dashed horizontal lines, but the caption describes a ‘Right’ panel for final accuracy comparison which is missing;

the current layout conflates baselines with training curves without a clear visual distinction for the ‘Right’ panel mentioned.

- **[writing]** Figure 2: The caption ‘GRPO HPA’ is insufficient; it fails to define the y-axis metric (‘Human Preference Accuracy’) or the x-axis (‘Step’), and does not explain the difference between the two plotted series (‘GRPO’ vs ‘GRPO (Parsing Text)’).
- **[science]** Figure 2: The x-axis tick labels are poorly formatted and illegible, with ‘1000’ and ‘1250’ overlapping (‘10001250’), which obscures the scale and data points in that region.
- **[science]** Figure 3: The plot displays two overlapping curves (dark blue and light blue) but lacks a legend or caption definition to distinguish the methods being compared, making the data uninterpretable.
- **[writing]** Figure 3: The caption ‘Text–Image Alignment’ is too brief to explain the axes (‘Reward Score’, ‘RL Iteration’) or the experimental context shown in the plot.
- **[writing]** Figure 4: The caption contains a grammatical error and missing term in the phrase ‘and -guided optimization’, failing to specify the method name (e.g., RL) that corresponds to the ‘RL’ label in the image.
- **[science]** Figure 4: The image displays two different female subjects for the SFT and RL columns respectively, rather than generating the same subject from the same prompt, which invalidates the visual comparison of the optimization method’s effect.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first point of use, creating barriers for non-specialist readers. Specifically, the term “VLM” (Vision-Language Model) appears frequently in the Abstract and Introduction without definition. Similarly, “OPD” (On-Policy Distillation) is introduced in the Introduction and Section 3.3 without a full expansion, assuming the reader already knows the specific distillation paradigm. In Section 5.3, the “GSB” metric is introduced without spelling out “Good-Same-Bad” first. Furthermore, statistical metrics like “PLCC” and “SRCC” are used in Section 4.1 without their full names (Pearson Linear Correlation Coefficient and Spearman Rank Correlation Coefficient), and “RL” is used in Section 5 without expansion. While these terms are standard in the immediate sub-field, the paper’s claim to represent a generalizable framework suggests a need for broader accessibility. The authors should define every acronym upon its first appearance in the text and consider replacing dense jargon with plainer descriptions where possible to ensure the core contribution is understood by a wider audience.

Action items.

- **[writing]** Define the acronym ‘VLM’ (Vision-Language Model) at its first occurrence in the Abstract or Introduction. It is currently used without definition, assuming reader familiarity.
- **[writing]** Replace the acronym ‘OPD’ (On-Policy Distillation) with the full term ‘on-policy distillation’ on first use in the Introduction and Section 3.3, or ensure it is explicitly defined before use.
- **[writing]** Replace the acronym ‘GSB’ (Good-Same-Bad) with the full phrase ‘Good-Same-Bad metric’ or ‘net preference score’ in Section 5.3 to avoid ambiguity for non-specialist readers.
- **[writing]** Replace the acronym ‘PLCC’ (Pearson Linear Correlation Coefficient) and ‘SRCC’ (Spearman Rank Correlation Coefficient) with their full names upon first introduction in Section 4.1 to improve accessibility.
- **[writing]** Replace the acronym ‘RL’ (Reinforcement Learning) with the full term ‘reinforcement learning’ at its first mention in Section 5 to ensure clarity for readers from adjacent fields.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The logical consistency of the paper is generally strong, with a clear narrative connecting the problem of scalar reward compression to the proposed teacher-student solution. However, there are specific areas where the causal claims and mathematical definitions require tighter alignment to fully support the conclusions.

First, in Section 3.1, the definition of the pairwise loss $\mathcal{L}^{\text{pw}}$ (Eq. 10) presents a potential inconsistency. The text states that pairwise supervision is applied across “all $G \times G$ cross-side sampled output pairs.” The summation in Eq. 10 iterates over $j \in \{w, l\}$, $i = 1 \dots G$, and $k = 1 \dots G$, resulting in $2 \times G^2$ terms. The normalization factor is $1/(2G^2)$. While mathematically this averages over the total terms, the text’s phrasing “applied across all $G \times G \dots$ pairs” might mislead a reader to expect a normalization of $1/G^2$ if they interpret the j summation as a separate aggregation step rather than part of the total count. Clarifying whether the “pairwise” nature refers to the w/l pairing or the i/k pairing in the normalization logic would prevent ambiguity.

Second, the central claim that the student “internalizes reasoning” (Abstract, Introduction) relies on the premise that the teacher’s score distribution q_T is a sufficient statistic for the reasoning process ρ_T . The distillation objective (Eq. 14) minimizes the KL divergence between q_T and q_ϕ . Logically, this only guarantees that the student matches the *output distribution* of the teacher, not that it has internalized the *reasoning mechanism*. The paper asserts that the distribution captures the “distributional effect of that reasoning,” but it does not provide evidence (e.g., an ablation where the teacher’s reasoning is removed or randomized to show the

distribution degrades) to prove that the distribution is causally dependent on the reasoning trace rather than just a high-capacity mapping from input to score. Without this, the claim of “internalizing reasoning” is a strong interpretation of a standard distribution-matching objective.

Third, in the analysis of the 9B model results (Section 5.2), the authors attribute the 9B GRPO model’s inferior performance to the “weaker reasoning ability of the 9B model.” This causal attribution is logically confounded. The 27B teacher uses GDSO (which includes direct score-gap supervision $\mathcal{L}^{\text{pw}}$), while the 9B GRPO baseline uses standard GRPO without this specific supervision. The performance gap could equally be caused by the absence of the direct supervision signal rather than the model size’s reasoning capacity. To support the claim that reasoning ability is the limiting factor, the authors should ideally compare a 9B model trained with GDSO against a 9B model trained with GRPO, isolating the training objective from the model capacity.

Action items.

- **[science]** In Section 3.1 (Eq. 10), the pairwise loss L_{pw} sums over G^2 pairs, but the normalization factor $1/(2G^2)$ implies an average over $2G^2$ terms. The text states supervision is applied across ‘all $G \times G$ cross-side sampled output pairs’ (G^2 pairs), yet the denominator includes a factor of 2. Clarify if the factor of 2 accounts for the symmetric summation over j in $\{w, l\}$ or if the normalization is inconsistent with the stated summation scope.
- **[science]** The claim that the student ‘internalizes reasoning’ relies on the assumption that the teacher’s distribution is a sufficient statistic for the reasoning process. The distillation objective (Eq. 14) only matches output distributions, not the reasoning mechanism. Provide evidence that the distribution is causally dependent on reasoning, not just a correlated output.
- **[science]** In Section 5.2, the paper attributes the 9B GRPO model’s lower performance to ‘weaker reasoning ability’. However, the 9B GRPO model lacks the direct score-gap supervision (L_{pw}) present in the 27B GDSO teacher. The performance gap could logically be attributed to the absence of this specific supervision signal rather than the model’s inherent reasoning capacity.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that slightly overreach the empirical evidence provided, particularly regarding the magnitude of performance gains and the nature of the distillation process.

First, the abstract and introduction repeatedly state that the 9B student “closely matches” the 27B teacher. While the Human Preference Accuracy (HPA) difference is small (89.6% vs

88.6%), a 1.0% absolute gap on a held-out test set with a massive 18B parameter disparity is non-trivial. Describing this as “closely matching” risks implying functional parity. A more precise characterization, such as “approaching” or “nearly matching,” would better reflect the remaining performance ceiling without overstating the efficiency of the distillation.

Second, the claim of a “41.3% net human-preference improvement” (Abstract, Section 5.3) is a misinterpretation of the Good-Same-Bad (GSB) metric. The GSB score is defined as $(G - B)/(G + S + B)$, which ranges from -1 to 1. A score of 0.413 indicates a net preference, but it does not equate to a 41.3% *improvement* in image quality or a 41.3% increase in the number of preferred images relative to the baseline. This phrasing conflates a normalized difference score with a relative percentage gain, which is an over-claim of the result’s magnitude.

Finally, the core contribution is framed as “internalizing reasoning” (Abstract, Section 3.2). However, the distillation objective (Eq. 11) minimizes the KL divergence between the student’s distribution and the teacher’s distribution *conditioned on the teacher’s reasoning trace*. The student is trained to predict the *outcome* of the teacher’s reasoning, not necessarily to replicate the reasoning process itself. While the student achieves high performance, the paper does not provide mechanistic evidence (e.g., probing internal states) that the student has actually “internalized” the reasoning logic. The claim should be tempered to reflect that the student internalizes the *judgment* derived from reasoning, rather than the reasoning process itself.

Action items.

- **[writing]** The claim that the 9B student ‘closely matches’ the 27B teacher (Abstract, Intro) is an over-interpretation of the 1.0% HPA gap (89.6% vs 88.6%). Given the 18B parameter difference, this gap is statistically significant and should be framed as ‘competitive’ or ‘nearly matching’ rather than ‘closely matching’ to avoid implying parity where a performance ceiling likely exists.
- **[writing]** The assertion that the method yields a ‘41.3% net human-preference improvement’ (Abstract, Sec 5.3) overstates the result. The GSB metric (Eq. 14) is a normalized difference score, not a direct percentage improvement in quality. Reporting this as a ‘41.3% improvement’ implies a relative gain in image quality that the metric does not strictly support; it should be reported as a GSB score of 0.413.
- **[writing]** The paper claims the student ‘internalizes’ reasoning without explicit chains (Abstract, Sec 3.2), yet the distillation target is the teacher’s distribution conditioned on the teacher’s reasoning trace. The paper does not provide evidence that the student’s internal representations actually encode the *reasoning* logic, only that it mimics the *output distribution*. Claiming ‘internalization of reasoning’ is an overreach; ‘internalization of reasoning-conditioned judgment’ is more accurate.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a novel framework for reward modeling in text-to-image generation, focusing on internalizing reasoning into score distributions. From a safety and ethics perspective, the work addresses the subjective nature of human preference, which is a positive step towards more nuanced alignment. However, several critical ethical and safety disclosures are missing or insufficiently detailed.

First, regarding human data collection, Section 3 (“Annotation and Datasets”) describes a workflow where annotators score images based on a rubric. While the process is detailed, there is no mention of Institutional Review Board (IRB) approval or ethical oversight. Given that human annotators are involved in generating training signals, standard ethical guidelines require explicit confirmation of IRB approval, informed consent from annotators, and measures taken to protect their privacy and well-being. The absence of this information makes it difficult to assess the ethical compliance of the data collection process.

Second, the paper does not adequately address potential dual-use risks. The proposed method optimizes text-to-image generation models to align with human preferences. While the goal is to improve quality, such systems can be misused to generate harmful, biased, or deceptive content (e.g., deepfakes, hate speech imagery, or non-consensual intimate imagery). The authors should include a dedicated discussion on these risks, the potential for reward hacking or over-optimization leading to unintended behaviors, and any specific safeguards or filtering mechanisms implemented to prevent misuse.

Third, the data provenance section mentions using “real-world prompts from users” and “internal captions.” The paper needs to clarify the data usage policies regarding these user-generated prompts. Specifically, did the authors obtain consent from users to use their prompts for training? How was personally identifiable information (PII) handled? Without this transparency, there are concerns regarding data privacy and the ethical use of user data.

Finally, while the paper mentions “quality-control annotators” and thresholds for accuracy, it does not discuss the potential for bias in the annotation process itself. If the rubric or the annotators’ judgments reflect societal biases, the resulting reward model could perpetuate or amplify these biases in the generated images. A brief discussion on bias mitigation in the annotation and training pipeline would strengthen the ethical standing of the work.

In summary, while the technical contribution is significant, the paper requires revisions to address these ethical and safety concerns, particularly regarding human subject research compliance, dual-use risks, and data privacy.

Action items.

- **[science]** The paper describes an annotation workflow involving human raters scoring images on subjective dimensions (Section 3). However, it lacks explicit details on IRB approval, informed consent procedures, or how annotator privacy and data protection were handled. This is a critical omission for reproducibility and ethical compliance.
- **[science]** The methodology involves training reward models to optimize text-to-image generation based on human preferences. The paper should explicitly discuss potential dual-

use risks, such as the model being used to generate harmful, biased, or deceptive content, and outline any mitigation strategies employed during training or deployment.

- **[science]** The dataset construction relies on ‘internally annotated’ data and ‘real-world prompts from users’ (Section 3). The authors must clarify the data usage policies, specifically whether user consent was obtained for using their prompts in training data, and how personally identifiable information (PII) was handled or removed.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence presented is generally robust in its experimental design, utilizing a teacher-student framework with clear baselines (SFT, GRPO, RewardDance, OPD) and multiple metrics (PLCC, SRCC, HPA, Margin HPA). The use of an internally annotated dataset with a detailed rubric and quality control workflow (Section 2) strengthens the validity of the ground truth. The ablation studies on score-distribution decoding (Figure 4) and the comparison with on-policy distillation (Table 3) provide compelling evidence for the specific design choices of the proposed method.

However, the statistical rigor regarding the final human evaluation of the text-to-image optimization (Section 5.3) is insufficient. The claim of a “41.3% net human-preference improvement” is a strong central result, yet the manuscript fails to report the underlying counts (G, S, B) or any measure of uncertainty (standard error, confidence intervals). Given the sample size of 400 prompts, the margin of error is non-negligible, and without these statistics, it is impossible to determine if the improvement is statistically significant or if the result could be attributed to random variance or annotator bias.

Furthermore, the teacher training objective (Eq. 12) is a hybrid of policy-gradient and direct supervised losses. While the authors argue this accelerates calibration, there is no ablation study isolating the effect of the direct supervised terms ($\alpha_{pt}\mathcal{L}_{CE}^{pt} + \alpha_{pw}\mathcal{L}^{pw}$) from the policy-gradient component. It is plausible that the performance gains are primarily driven by the strong direct supervision rather than the novel distributional policy optimization, which would weaken the claim that the specific GDSO mechanism is the primary driver of success.

Finally, the comparison between the 9B student and the 27B teacher relies on a small absolute difference in HPA (88.6% vs 89.6%). Without a statistical significance test (e.g., a bootstrap test on the HPA metric), the assertion that the student “closely matches” the teacher is not fully supported by the evidence provided. The paper should include these statistical validations to ensure the central claims are robust to plausible alternative explanations regarding sample size and effect magnitude.

Action items.

- **[science]** The paper claims a 41.3% net human-preference improvement (GSB) for the optimized generator but does not report the standard error, confidence intervals, or the raw counts (G, S, B) for the 400-prompt evaluation. Without these statistics, the significance of the effect size and the robustness of the claim against random variation cannot be assessed.
- **[science]** The teacher training combines policy-gradient (GRPO) with direct supervised losses (CE and pairwise gap). The paper does not provide an ablation study isolating the contribution of the direct supervised terms versus the policy-gradient term. It is unclear if the performance gains stem from the distributional objective or simply from the strong supervision signal, which risks conflating the proposed method’s novelty with standard supervised fine-tuning benefits.
- **[science]** The student distillation (RISD) achieves 88.6% HPA, nearly matching the 27B teacher (89.6%). However, the paper lacks a statistical test (e.g., bootstrap or t-test) to confirm that this small gap is not statistically significant. Without this, the claim that the student ‘closely matches’ the teacher remains anecdotal rather than empirically proven.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in the paper is generally sound in its conceptual approach, particularly the shift from scalar rewards to score distributions to capture human uncertainty. However, several specific statistical reporting and validation gaps need to be addressed to ensure the robustness of the claims.

First, regarding the annotation process described in Section 3, the authors state that the ground-truth score distribution is computed by “dropping the highest and lowest scores before aggregation.” While this is a common heuristic to reduce outlier noise, the paper lacks a statistical justification for this specific trimming rule. Without reporting the number of annotators per sample or conducting a sensitivity analysis (e.g., comparing trimmed vs. untrimmed means/-variances), it is unclear if this procedure introduces bias, especially if the human preference distribution is not symmetric. The authors should clarify the sample size per item and justify the trimming threshold statistically.

Second, the human evaluation results in Section 6.3 report a “net GSB improvement of 41.3%.” This is a point estimate derived from 400 prompts. The manuscript fails to provide a confidence interval (e.g., 95% CI) or a p-value to assess the statistical significance of this improvement over the SFT baseline. Given the binary nature of the pairwise comparison (Good vs. Bad), a simple binomial test or McNemar’s test should be performed to confirm that the observed gain is statistically significant and not a result of random variation.

Third, the ablation study in Figure 4 (and the accompanying text) claims that using the

distribution expectation yields “consistently” better results than parsing text. However, the figures do not display error bars, and the text does not mention whether the experiments were repeated with multiple random seeds. To support the claim of “consistent” improvement, the authors must report the standard error or standard deviation across seeds and, ideally, the p-values of the differences. Without this, the observed gains could be attributed to variance in the training process rather than the method itself.

Finally, a minor but notable typo exists in Table 1 (9B RewardDance row), where the SRCC is listed as “6175” instead of “0.6175”. While likely a formatting error, such inaccuracies in statistical reporting can erode trust in the precision of the results. All numerical values should be double-checked for decimal placement and consistent significant figures.

Addressing these points—specifically the statistical validation of the human evaluation and the robustness of the ablation results—will significantly strengthen the empirical claims of the paper.

Action items.

- **[science]** The annotation workflow (Sec 3) states that ground-truth distributions are computed by dropping the highest and lowest scores. However, the statistical validity of this outlier removal is not justified with a sensitivity analysis or a test for normality. If the underlying distribution of human scores is bimodal or skewed, trimming extremes may bias the mean and variance estimates used for training. Please report the number of annotators per sample and justify the trimming rule statistically.
- **[writing]** In Table 1, the 9B RewardDance row lists an SRCC of ‘6175’ (likely a typo for 0.6175). While likely a formatting error, such precision issues in statistical reporting undermine confidence in the reported metrics. Please verify all decimal placements and ensure consistent significant figures across all tables.
- **[science]** The human evaluation in Sec 6.3 reports a net GSB improvement of 41.3% based on 400 prompts. The paper does not provide a confidence interval (e.g., Wilson score interval) or a p-value for this difference against the SFT baseline. Given the binary nature of the ‘Good’ vs ‘Bad’ comparison, a statistical significance test (e.g., McNemar’s test or a binomial test) is required to confirm the improvement is not due to chance.
- **[science]** The ablation study in Fig 4 compares ‘Parsing Text’ vs ‘Distribution Expectation’. The text claims the latter provides ‘denser supervision,’ but the statistical significance of the observed gains in HPA and Margin HPA is not quantified. Please include error bars (e.g., standard error over 3 random seeds) or report p-values to ensure the improvements are robust and not artifacts of random initialization.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of academic writing, with a clear logical flow from the problem statement to the proposed solution and experimental validation. The abstract effectively summarizes the core contributions, and the introduction successfully establishes the tension between reasoning quality and inference efficiency. The prose is generally precise, and the mathematical notation is consistent throughout the document.

However, there are a few areas where sentence structure and clarity could be improved to enhance readability. In Section 3, the description of the score distribution decoding mechanism uses the verb “decode” without fully clarifying the underlying token-generation process, which may cause slight ambiguity for readers unfamiliar with the specific Q-Align implementation details. Additionally, in Section 4.2, the comparison of RewardDance’s performance metrics is presented in a single, information-dense sentence; breaking this down would allow the reader to better digest the trade-off between preference accuracy and score calibration. Finally, in Section 5, a long sentence describing the reward backpropagation scheme contains a relative clause that could be misinterpreted as modifying the wrong noun phrase. A minor restructuring here would eliminate this potential ambiguity.

Overall, the writing is strong and effectively communicates the technical contributions. Addressing these specific points will further polish the manuscript.

Action items.

- **[writing]** In Section 3, clarify the term ‘decode’ regarding the score distribution. It is unclear if this refers to token generation or logit derivation. Ensure the mechanism matches the description.
- **[writing]** In Section 4.2, split the dense sentence comparing RewardDance’s HPA and calibration metrics into two sentences to improve readability and clarity.
- **[writing]** In Section 5, rephrase the sentence about the ReFL-style scheme. The relative clause ‘which is closely related...’ ambiguously modifies ‘optimization’ instead of ‘scheme’. Clarify the subject.